

# Pros/Cons of WiFi

## ➤ Pros

- Mobility
- Simple Installation
- Inexpensive

## ➤ Cons

- Security
- Low bandwidth (compared to wired networks)
- Interference with other devices (e.g., Bluetooth or even microwave oven!)

# Mobile Supports May Come From?

## ➤ Link layer:

- Examples: IEEE 802.11r Fast Roaming (not widely used) or cellular networks
- Allows to handover among access points
- Pro: No changes to clients (mobile nodes) needed
- Con: Works only for specific wireless networks

## ➤ Network layer:

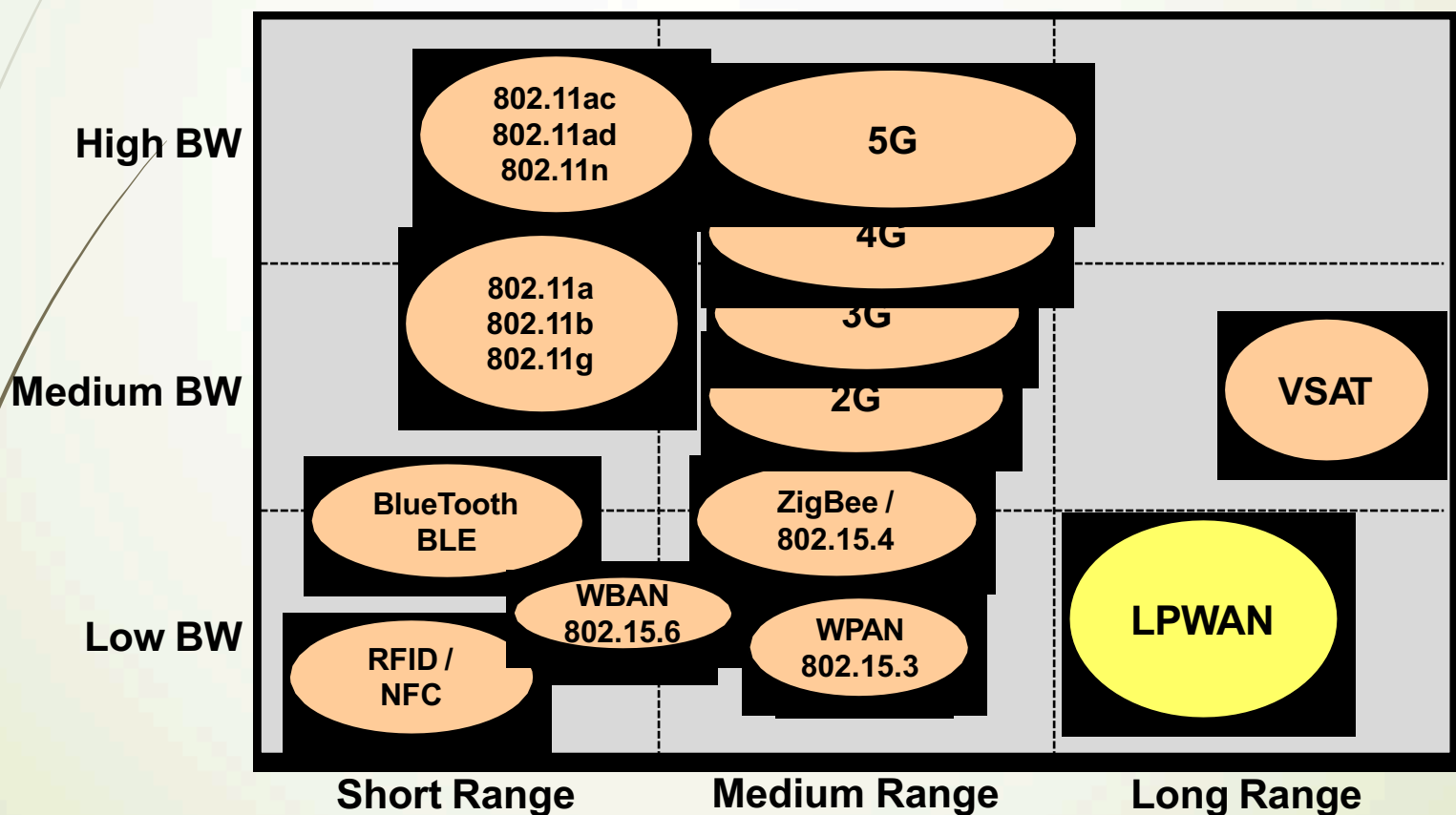
- Examples: Mobile IP, Proxy MIP
- Transparent to transport protocols; thus applications are unaware of changes of network attachment (handover)
- Pro: Works for different wireless technologies
- Con: Changes in OS for mobile nodes required

## ➤ Application layer:

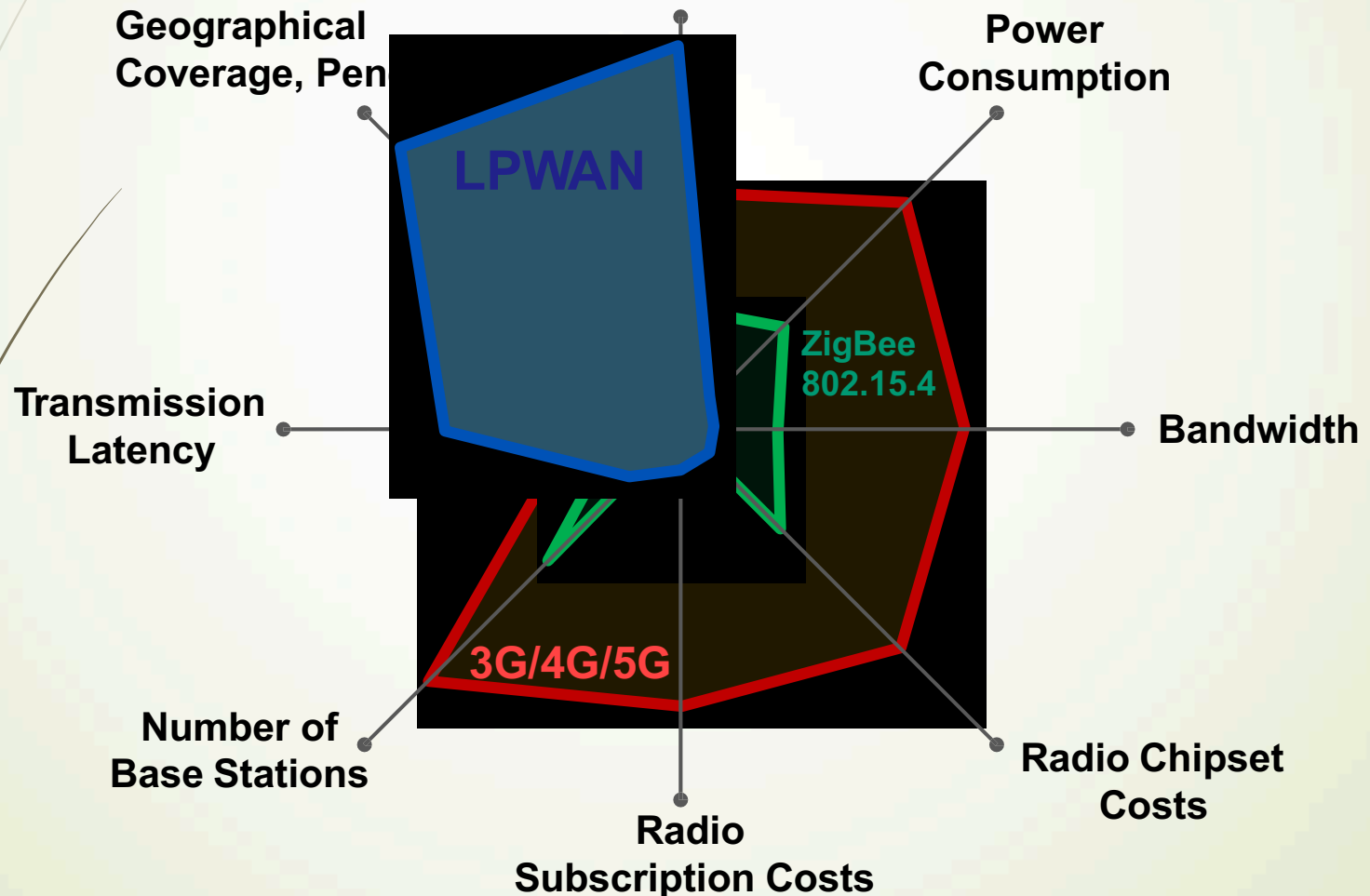
- Examples: SIP registrations, DNS/dynDNS
- Pro: No changes to clients (mobile nodes) needed
- Con: Disruptive (an open connection will be dropped), thus only suited for quasi-static attachment to network using DHCP or PPPoE for obtaining IP address, e.g. once a day)

# LPWAN – Low Power Wide Area Network

Become popular due to IoT and M2M applications



# LPWANs Have Unique Requirements



# Targets of LPWANs

| Requirement                                   | Target                                                                                                                         |
|-----------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------|
| Long range                                    | 5 – 40km in the open field                                                                                                     |
| Ultra low power                               | Battery lifetime of 10 years                                                                                                   |
| Throughput                                    | A few hundred bps or less                                                                                                      |
| Radio chipset costs                           | \$2 or less                                                                                                                    |
| Radio subscription costs                      | \$1 per device and year                                                                                                        |
| Transmission latency                          | Not a primary requirement for LPWAN                                                                                            |
| Required number of base stations for coverage | Very low. LPWAN base stations are able to serve thousands of devices.                                                          |
| Geographic coverage, penetration              | Excellent coverage also in remote and rural areas. Good in building and in-ground penetration (e.g. for reading power meters). |

# LPWAN over the ISM Band

| LPWAN Technology | Standard / Specification                                           | Range                                           | Spectrum                                                               |
|------------------|--------------------------------------------------------------------|-------------------------------------------------|------------------------------------------------------------------------|
| ETSI LTN         | ETSI GS LTN 001 - 003                                              | 40 km in open field                             | Any unlicensed spectrum such as ISM (433MHz, 868MHz, 2.4GHz)           |
| LoRaWAN          | LoRa Alliance LoRaWAN                                              | 2-5km in urban areas<br><15km in suburban areas | Any unlicensed spectrum<br>868MHz (Eu)<br>915MHz (US)<br>433MHz (Asia) |
| Weightless-N     | Weightless SIG                                                     | <5km in urban areas 20-30km in rural areas      | 800MHz – 1GHz (ISM)                                                    |
| RPMA             | Proprietary (On-Ramp Wireless), planned to become an IEEE standard | <65km line of sight<br><20km non line of sight  | 2.4GHz                                                                 |

# Questions

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